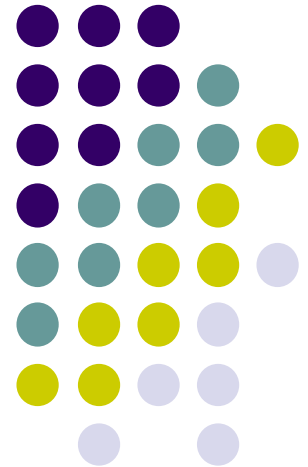


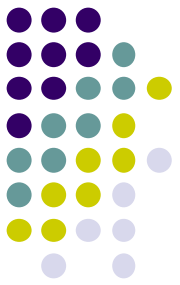
WIX1002

Fundamentals of Programming

Chapter 3

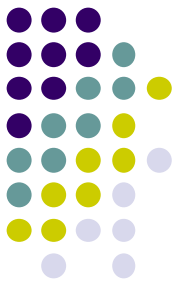
Flow of Control (Selection)





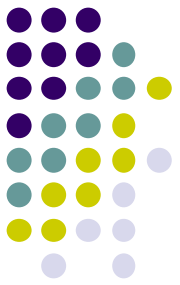
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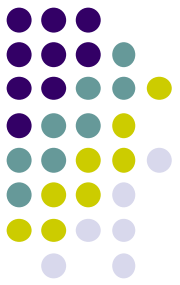
Introduction

- Flow control is to used to specify the order of the statements to be executed.
- Program can be written in three types of flow control namely the **sequence, selection and repetition**.
- When the statements are executed **one after the other** in order, it is called the **sequence flow**.
- A **selection flow** chooses among **alternative courses of action**.
- A **repetition flow** specifies that an action is to be **repeated while some condition remains true**.



Introduction

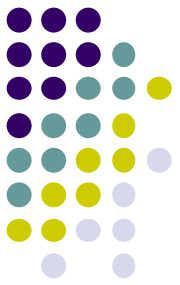
- Computer programs often need to make decisions, taking different actions depending on the condition.
- In Java, **if** and **switch** statement are used to carry out the decision.
- The statement is controlled by the **boolean expression**.
- **Relational/Conditional operator** can be used in the boolean expression.
- If there is more than one **constraint/condition** in the decision making, the **logical operators** are used to merge multiple constraint/conditions.



Relational Operator

- The relational operator tests the relationship between two values.

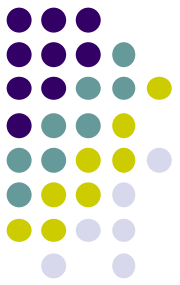
Operator	Description	Examples
==	Equal	a==b
!=	Not Equal	c!=d
>	Greater than	x>y
>=	Greater than or equal	x>=y
<	Less than	a<b
<=	Less than or equal	a<=b



Logical Operator

- The logical operator is used to create complex Boolean expression by merging multiple constraints/conditions.

Operator	Description	Examples
&&	AND (true && true is true, others false)	a==b && c==d
	OR (false && false is false, others true)	c!=d a<b
!	NOT (!false is true. !true is false)	!(x>y)



if

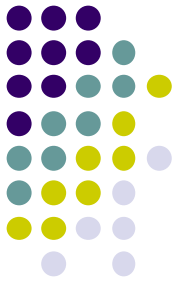
- if statement is used to implement a decision. It consists of condition and body.
- If the condition is true, the body of the statement is executed.

```
if ( condition1 )  
    statement 1;
```

```
// use brace {  
if ( condition1 ) {  
    statement 1;  
    statement 2;  
}
```

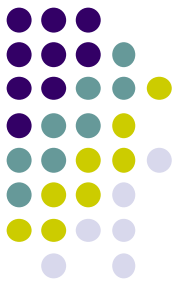
more than 1 statements

if



```
if (number > 0)
    System.out.println("The number is positive");
```

```
if (result < 50) {
    System.out.println("You did not pass");
    System.out.println("Try harder next time");
}
```

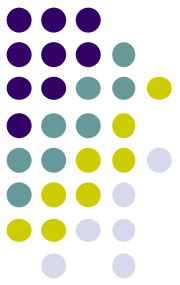
if-else

- if-else statement chooses between two alternative statements based on the condition or boolean expression

```
if ( condition1 )  
    statement 1;  
else  
    statement 2;
```

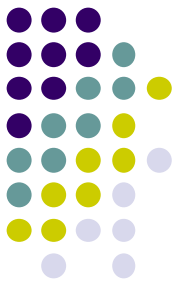
```
if (myScore > yourScore)  
    System.out.println("I Win!");  
else  
    System.out.println("You Win!");
```

if-else



```
// use brace {  
if ( condition1 ) {  
    statement 1;  
    statement 2;  
}  
else {  
    statement 3;  
    statement 4;  
}
```

more than 1 statements



if-else

- String Comparison
 - When testing string for equality, **DO NOT USE ==** operator. Use **equals** or **equalsIgnoreCase**.
 - `String.equals(other_string)`
 - `String.equalsIgnoreCase(other_string)`

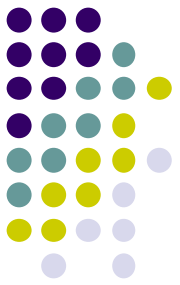
```
String s1, s2;
```

```
if (s1.equals(s2))
```

```
    System.out.println("They are equal strings.");
```

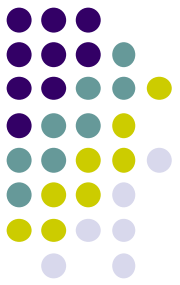
```
else
```

```
    System.out.println("They are not equal strings.");
```



if-else

- **Alphabetical Order**
 - Lexicographic ordering is used to order alphabet according to ASCII ordering. Use **compareTo** and **compareToIgnoreCase**.
 - `String.compareTo (other_string)`
 - `String.compareToIgnoreCase(other_string)`
 - `s1.compareTo(s2)`
 - Return **negative** value if **s1 comes before s2**.
 - Return **positive** value if **s2 comes before s1**.
 - Return zero if s1 is equal to s2.



Multiway if-else

- Multiway if-else statement is the if-else statement nested inside the if-else statement

```
if ( condition1 )
```

```
    statement 1;
```

```
else if ( condition2 )
```

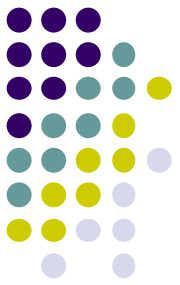
```
    statement 2;
```

```
else if ( condition3 )
```

```
    statement 3;
```

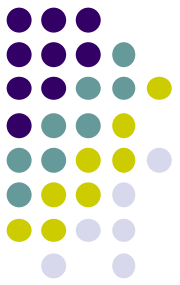
```
else
```

```
    statement 4;
```



Multiway if-else

```
if (myScore > yourScore) {  
    System.out.println("I Win!");  
}  
else if (myScore < yourScore) {  
    System.out.println("You Win!");  
}  
else {  
    System.out.println("Tie!");  
}
```



switch

- switch statement can be used to represent multiway if-else statement.

```
switch ( variable ) {
```

```
case value1:
```

```
    statement 1;
```

```
    break;
```

```
case value2:
```

```
    statement 2;
```

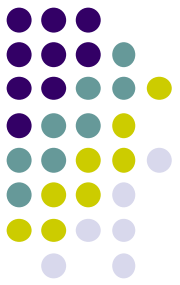
```
    break;
```

```
default:
```

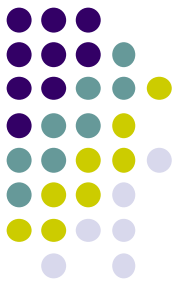
```
    statement 3;
```

```
}
```

switch

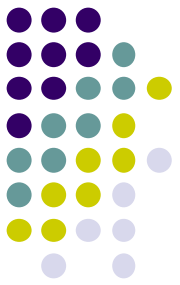


```
switch (number) {  
    case 1:  
        System.out.println("Satu");  
        break;  
    case 2:  
        System.out.println("Dua");  
        break;  
    case 3:  
        System.out.println("Tiga");  
        break;  
    default:  
        System.out.println("This program accepts the number from 1 to 3  
only");  
}
```

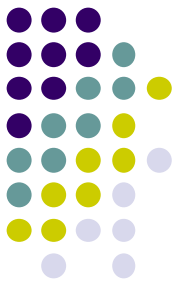
Ternary Operator

- The ternary operator `? :` is similar to if-else statement
- **`condition1 ? statement1 : statement2`**
 - If the condition1 is true, the statement1 will be executed.
 - If the condition1 is false, the statement2 will be executed.
 - `y = x >= 0 ? x : -x;`



Common Error

- if number > 0
 - System.out.println("No bracket");
- if (x > 5 * (y - z)
 - System.out.println("Miss one bracket");
- if (0 <= mark <= 100)
 - System.out.println("Invalid syntax");
- if (choice = 'Q')
 - System.out.println("Equal sign error");
- if (age > 21 and mark > 80)
 - System.out.println("and should be &&");



Common Error

- ```
switch (monthNum) {
 case 1:
 System.out.println("January");
 case 2:
 System.out.println("No break");
}
```

